# Akshar Raikanti

(925) 416 - 9570 | akshar.raikanti@gmail.com | linkedin.com/in/aksharraikanti/

## EDUCATION

**Purdue University** **West Lafayette, IN**
***B.S., Computer Science*** **December 2026**
***B.S., Artificial Intelligence*** **GPA: 3.62**
**Relevant Courses:** Analysis of Algorithms, Systems Programming, Artificial Intelligence, Data Structures and Algorithms

**Skills:** Python, Java, C, C++, C#/.NET, SQL, R, JavaScript, TensorFlow, PyTorch, Relational Databases, NoSQL, Azure, Google Cloud (GCP), AWS, LangChain, LangGraph, Agentic AI, RAG, Vector Database Search, ML Training, Docker, CI/CD, OpenShift, Jira, Apache, Git

## PROFESSIONAL EXPERIENCE

**Tesla** **Fremont, CA**
*Software Engineer Intern* **`September 2025 – December 2025**

- Built a Health Check AI Agent that correlates logs, runs post deployment checks to verify application; detects early warning patterns using RAG on historical data, catching potential issues with 85% accuracy.
- Engineered a multimodal AI pipeline to digitize 10,000+ legacy PDFs, utilizing dual-model inference (image analysis + semantic validation) to automatically convert static work instructions into proprietary data models.
- Implemented a full-stack application (Python/Golang/Angular) to facilitate human-in-the-loop review, creating an interface that flags low-confidence AI predictions for mandatory engineering approval to ensure 100% process safety.
- Architected a microservices-based orchestrator agent within Microsoft Teams, implementing SSO to secure inter-agent communication and enable granular, user-authenticated actions for hardware logistics and database management.

**Textron** **Hunt Valley, MD**
*Agentic AI Software Engineer Intern* **August 2025 – September 2025**

- Built an Agentic AI RAG workflow in C# + Python that ingests docs; tokenizes, generates and indexes OpenAI embeddings; decision agent reaches the right team with context leading to 40% fewer misroutes using Microsoft Semantic Kernel.
- Trained a bid/no-bid decision engine (RAG on Textron data via FAISS + GPT-4o) that 3x screening throughput and cut manual review time by 50%, while including win-probability scoring on high priority opportunities.
- Collaborated with Microsoft to build a Databricks ingestion/embedding pipeline using Logic Apps for orchestration and Microsoft AI Foundry for model lifecycle integrated with Azure Data Factory to deliver low-latency agent knowledge lookup.

**Kohl's** **San Francisco, CA**
*Software Engineer Intern* **June 2025 – August 2025**

- Engineered a scalable high-performance backend for an ordering platform with RESTful API, leveraging Java Spring Boot, MongoDB, Google Cloud Spanner cloud databases, achieving 10% increase in responsiveness, 5% reduction in load times.
- Developed intuitive frontend interfaces using Angular (TypeScript), enhancing Kohl's order management; modernized CI/CD pipelines using Auto DevOps and container orchestration via Red Hat OpenShift (Kubernetes) accelerating deploys by 5%.

**TransSIGHT** **San Francisco, CA**
*Data Engineering and Machine Learning Intern* **June 2024 – August 2024**

- Built and deployed predictive machine learning transit-demand models (Random Forest, XGBoost; scikit-learn, PyTorch) and designed an automated end-to-end data pipeline using Apache Airflow, AWS S3/Lambda, raising forecast accuracy by 20%.

## PROJECTS

**Agentic AI Workflow**

- Engineered an agentic Web Search + Retrieval-Augmented Generation (RAG) system on Nvidia NIM (Llama-3.1-8B) using AIQ Toolkit, LangGraph (ReAct/tool-calling), and LangChain; registered a custom Tavily Internet Search Tool and a local-filesystem RAG pipeline (document chunking, embeddings, vector search via FAISS) to deliver fast multi sourced answers to user requests.
- Productionized the agent as a FastAPI + LangGraph microservice (MCP-compatible) with Redis caching, request dedup, and streaming.

**Local LLM**

- Built and deployed a local LLM by fine-tuning a T5 Seq2Seq model on Q&A using Hugging Face Transformers and PyTorch; evaluated with ROUGE metrics, optimized via BitsAndBytes 8-bit quantization, benchmarked latency with custom scripts; served with a secured FastAPI.

**Pantry Tracker App**

- Built a responsive web app using React and Next.js for real-time pantry item tracking; implemented secure user authentication via Firebase and automated item addition using computer vision with OpenCV.

## RESEARCH EXPERIENCE

**BASF** – Agricultural Chemical Solutions **West Lafayette, IN**
*Machine Learning Research Intern* **August 2023 – May 2024**

- Developed statistical models and predictive analytics using Python, scikit-learn, XGBoost, and Tableau, enhancing competitor market-share estimation accuracy and delivering actionable benchmarking insights to senior management leading to stronger business strategy decisions.